Domain of x : Set of all students of the class

Domain of y : Set of all classes of the school

Domain of z : Set of all departments of the school

$$\forall x \exists y \exists z \left((y \neq z) \land T(x,y,Math) \land T(x,z,Math) \land \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$

$$\neg \forall x \exists y \exists z \left((y \neq z) \land T(x,y,Math) \land T(x,z,Math) \land \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right) \equiv \exists x \neg \exists y \exists z \left((y \neq z) \land T(x,y,Math) \land T(x,z,Math) \land \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$ (De Morgan's law of quantifiers)

$$\equiv \exists x \forall y \neg \exists z \left((y \neq z) \land T(x,y,Math) \land T(x,z,Math) \land \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$ (De Morgan's law of quantifiers)

$$\equiv \exists x \forall y \forall z \neg \left((y \neq z) \land T(x,y,Math) \land T(x,z,Math) \land \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$ (De Morgan's law of quantifiers)

$$\equiv \exists x \forall y \forall z \left(\neg (y \neq z) \lor \neg T(x,y,Math) \lor \neg T(x,z,Math) \lor \neg \forall w \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$ (1st De Morgan's law)

$$\equiv \exists x \forall y \forall z \left((y = z) \lor \neg T(x,y,Math) \lor \neg T(x,z,Math) \lor \exists w \neg \left(T(x,w,Math) \Rightarrow \left((w=y) \lor (w=z) \right) \right) \right)$$ (De Morgan's law of quantifiers)

$$\equiv \exists x \forall y \forall z \left((y = z) \lor \neg T(x,y,Math) \lor \neg T(x,z,Math) \lor \exists w \neg \left(\neg T(x,w,Math) \lor \left((w=y) \lor (w=z) \right) \right) \right)$$ (Conditional-disjunction equivalence)

$$\equiv \exists x \forall y \forall z \left((y = z) \lor \neg T(x,y,Math) \lor \neg T(x,z,Math) \lor \exists w \left(\neg (\neg T(x,w,Math)) \land \neg (w=y) \land \neg (w=z) \right) \right)$$ (2nd De Morgan's law)

$$\equiv \exists x \forall y \forall z \left((y = z) \lor \neg T(x,y,Math) \lor \neg T(x,z,Math) \lor \exists w \left(T(x,w,Math) \land (w \neq y) \land (w \neq z) \right) \right)$$ (Double negation Law)

There exists some student in the class such that for any two arbitrary Math department courses, the student has not taken atleast one of them or there exists atleast one other